A Three-Person Attendance Game: Payoff Definition

Players, states, and actions. There are three players, $N = \{1, 2, 3\}$, and every pair knows one another: the potential contact graph is K_3. At the start of a day, let $x_i \in \{0, 1\}$ denote player i's disease state, where 0 means susceptible (S) and 1 means infected (I). The example below fixes $x = (1, 0, 0)$, and all three players know this state. Each player simultaneously chooses $a_i \in \{A, H\}$, where A means attend and H means stay home. A pair meets on site exactly when both attend:

$$C_{ij}(a) = \mathbf{1}\{a_i = A\}\mathbf{1}\{a_j = A\}, \qquad i \neq j.$$

Staying home removes all contacts in this small network. There are no outside infections. Transmissions and recoveries use start-of-day states; new states take effect at the end of the day.

Probability of infection. For a susceptible player i, the number of infected neighbors who attend alongside i is

$$m_i(a, x) = \sum_{j \in N \setminus \{i\}} x_j C_{ij}(a).$$

Each active infected–susceptible edge independently transmits with probability β. The probability of *avoiding* infection on every such edge is therefore

$$\Pr(X_i' = 0 \mid x_i = 0, x, a) = \prod_{\substack{j \neq i \\ x_j C_{ij}(a) = 1}} (1 - \beta) = (1 - \beta)^{m_i(a,x)}.$$

Taking the complement gives

$$\Pr(X_i' = 1 \mid x_i = 0, x, a) = 1 - (1 - \beta)^{m_i(a,x)}.$$

An initially infected player transmits during the day and recovers at its end with probability γ. Thus, defining $q_i(a, x) = \Pr(X_i' = 1 \mid x, a)$,

$$q_i(a, x) = \begin{cases} 1 - (1 - \beta)^{m_i(a,x)}, & x_i = 0, \\ 1 - \gamma, & x_i = 1. \end{cases}$$

Realized and expected payoffs. Let $c_i > 0$ be player i's one-day cost of staying home (lost work or social activity), and let $L_i > 0$ be the cost of *ending the day infected*. Attendance has zero activity cost by normalization. If the end-of-day infection indicator is $X_i' \in \{0, 1\}$, realized payoff is

$$u_i(a, X_i') = -c_i \mathbf{1}\{a_i = H\} - L_i X_i'.$$

Before the random disease outcome is observed, the expected payoff is

$$\begin{aligned} U_i(a \mid x) &= \mathbb{E}[u_i(a, X_i') \mid x, a] \\ &= -c_i \mathbf{1}\{a_i = H\} - L_i \mathbb{E}[X_i' \mid x, a] \\ &= -c_i \mathbf{1}\{a_i = H\} - L_i \Pr(X_i' = 1 \mid x, a) \\ &= \boxed{-c_i \mathbf{1}\{a_i = H\} - L_i q_i(a, x)}. \end{aligned}$$

The third equality holds because the expectation of a 0–1 indicator equals the probability that it is 1.

Numerical example and best responses. Use $c_i = 0.2$, $L_i = 10$, $\beta = 0.04$, and $\gamma = 0.20$ for every player; these values illustrate the game and are not empirical estimates of the costs. With $x = (1, 0, 0)$, player 1's end-of-day infection probability is $1 - \gamma = 0.8$, regardless of attendance:

$$U_1(A \mid x) = -10(0.8) = -8, \qquad U_1(H \mid x) = -0.2 - 10(0.8) = -8.2.$$

Player 1 therefore prefers A. If player 1 chooses A, then each susceptible player $i \in \{2, 3\}$ faces one infected attending neighbor when also choosing A. Hence

$$\begin{aligned} m_i(A, a_1 = A, x) = 1, &\quad U_i(A, a_1 = A \mid x) = -10[1 - (1 - 0.04)^1] = -0.4, \\ m_i(H, a_1 = A, x) = 0, &\quad U_i(H, a_1 = A \mid x) = -0.2 - 10[1 - (1 - 0.04)^0] = -0.2. \end{aligned}$$

Both susceptible players prefer H when player 1 attends. Therefore (A, H, H) is a Nash equilibrium. Player 1's attendance raises others' infection risk without raising player 1's private infection risk: this is the externality the later policy model would need to address.